INCH-POUND

MIL-PRF-53013B
01 February 1999
SUPERSEDING
MIL-R-53013A(ME)
28 May 1992

PERFORMANCE SPECIFICATION

REPAIR KIT, 6-INCH HOSELINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a repair kit consisting of tools and materials necessary for permanent repair to the Inland Petroleum Distribution System (IPDS) 6-inch, lightweight, collapsible hose-line. The components are contained in a fitted storage chest.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-D/210, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4610

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SPECIFICATIONS

FEDERAL

A-A-50271	- Plate, Identification.
A-A-52557	- Fuel Oil, Diesel, for Posts, Camps and Stations.
A-A-52598	- Couplings, Clamp, Pipe; with Bolts and Synthetic-Rubber Gaskets for Grooved-End Pipe.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 910	- Standard Specification for Aviation Gasolines.
ASTM D 975	- Standard Specification for Diesel Fuel Oils.
ASTM D 1655	- Standard Specification for Aviation Turbine Fuels.
ASTM F 1166	- Standard Practice for Human Engineering Design for Marine Systems, Equipment and Facilities.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Description. The hose-line repair kit (hereinafter called “kit”) shall consist of the components described herein and be fitted in a storage chest.

3.2 Materials. Materials shall be selected by the contractor and shall be subject to all provisions of this specification. Materials selected shall meet or exceed the requirements specified herein and shall be sufficient to withstand the rigors of daily use in the applicable operating and storage environments to which the parts may be exposed, without suffering the effects of deterioration. Storage environment for the repair kit is - 40 °F to 160 °F and operating environment is -20 °F to 135 °F respectively with humidity of 0-100%.

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3.2.1 Identification of materials and finishes. The contractor shall identify and document the specific materials, material finish, or treatment for use with components and sub-components.

3.2.2 Recovered material. The offerer/contractor is encouraged to use recovered materials (see 6.6.2) to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.2.3 Deterioration prevention and control. The equipment shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration to which the materials are susceptible.

3.2.3.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

3.3 Repair kit parts list. The components comprising the repair kit and their required quantities are listed in table I.

TABLE I. 6-Inch Hose-Line Repair Kit Parts.

Qty.	Description	Ref No.
3	Coupling, clamp, 6 inch nom. size	A-A-52598
6	Gasket, coupling, 6 inch nom. size	A-A-52598
2	Clamp, hose, 6 inch	(figure 1)
18	Hose nipple assembly, grooved end, three-segment, 6 inch	(figure 2)
6	Hose repair assembly, grooved end, three-segment, 6 inch	(figure 3)
6	Adapter assembly, hose, double grooved end, three-segment, 6 inch	(figure 4)
1	Set, wrench, socket, 12 point, 1/2 drive	(3.6.1)
1	Shears, 8 inch nom. size.	(3.6.2)
1	Knife, rubber cutting, non- conductive handle, 8.00 in. × 1.00 in.	(3.6.3)

3.4 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.5 Hose clamp. The hose clamp shall consist of a base, fastener, frame, crosshead, and a screw and screw handle (pictured) or other means for lowering the crosshead and compressing the hose (see figure 1).

3.5.1 Base. Means shall be provided for latching and pivoting the vertical frame.

3.5.2 Fastener. The fastener shall be self-actuating and affixed to the base in such a manner that it may move freely. Means shall be provided for quickly engaging and locking the movable side of the frame.

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3.5.3 Frame. The frame shall be U-shaped. One side of the frame shall act as a hinge, mounted on the base, and the other shall have the capability of being securely engaged by the fastener. The inside of the frame shall have parallel tracks, free of burrs and fins, to act as guide for the crosshead. The crosshead shall slide over the entire length without binding. The frame shall have not less than 11.50 inch clearance between the legs of the "U" and shall provide not less than 6.50 inch clearance between the lower jaw and the crosshead when the frame is closed and the crosshead is in its uppermost position.

3.5.4 Crosshead. The top of the crosshead shall incorporate a method for quickly lowering the crosshead so that a hose can be compressed against the lower jaw and stop the flow of fluid through the hose without damaging the hose.

3.5.5 Handle. The handle shall move without binding in the screw and shall be equipped at each end with a means to prevent the handle from pulling out.

3.6 Tools.

3.6.1 Set, wrench, socket. The socket set shall provide an assortment of sockets which will as a minimum, fit all the bolts provided with the hose clamps and assemblies. In addition sockets shall be 12 point (double hexagon), 1/2 drive, regular length.

3.6.2 Shears. A pair of shears shall be provided for cutting IPDS 6-inch lightweight, collapsible hose line.

3.6.3 Knife. The rubber cutting knife shall have a non-conductive handle and a blade with at least 6 inches in length., suitable for cutting rubber hose. The handle shall be either impervious, or treated to minimize, impregnation by fuel and oil and designed to promote a positive gripping surface.

3.7 Storage chest. All components of the repair kit shall be fitted in a storage chest. The chest shall have a hinged cover, fastening devices, and handles. Bins, cradles, or blocking for components of the repair kit shall be provided to prevent damage during shipment or handling. A loading plan indicating the location of all items in the chest shall be provided. The loading plan shall be permanently attached to the inside of the cover.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article inspection is required, it shall be performed on one complete unit. The inspection shall include the examination 4.4 and the testing 4.5.

4.3 Conformance inspection. Conformance inspection shall include the examination 4.4.

4.4 Examination. Each repair kit shall be physically examined for compliance with the requirements specified in section 3. Any redesign or modification to comply with specified

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requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all examinations of performance, safety, human engineering, and dimensional requirements. Non-compliance with any specified requirement, or the presence of one or more defects shall constitute cause for rejection.

4.5 Testing. The repair kit shall be tested by using the hose clamp (see 3.5) to clamp an IPDS 6-inch, lightweight, collapsible hose-line and repairing it as follows:

1. Pressurize a plugged test hose segment to 200 psi minimum with water.
2. Place the hose clamp around the hose approximately 3 feet from the plugged end of the test hose segment.
3. Apply sufficient pressure with the hose clamp to the test hose to stop water flow.
4. After the hose has been compressed to stop fluid flow, cut the hose half-way between the clamp and the hose end-plug.
5. Any continuing flow (after initial draining) greater than 1 cup per minute shall constitute failure of this test.
6. Reattach the cut hose ends with a grooved end hose repair assembly (see figure 3).
7. Disengage the hose clamp.
8. The hose shall be checked for leakage after being re-pressurized and at half-hour intervals for the next 3 hours.
9. There shall be no damage to the hose at point of hose clamp engagement. Leakage of the hose at the repair site, and any permanent deformation of the hose clamp or hose hardware shall constitute failure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The kit is intended for use in making permanent repairs on the Inland Petroleum Distribution System (IPDS) 6-inch, lightweight, collapsible hose-line. The hose clamps should be used to crimp pressurized hoses on both sides of a damaged section and enable the repair crew to cut the damaged section out and install hose couplings or repair assemblies contained in the kit, as required, to effect a repair with minimum time and effort.

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6.2 Acquisition requirements. Acquisition documents will specify the following:

a. Title, number, and date of this specification
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2)
c. When first article equipment is not required or time frame for submission of first article (see 3.4)
d. Packaging requirements (see 5.1).

6.3 First article inspection. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Data Requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractors maintenance and operation manuals to be furnished with each repair kit.

6.5 Documents. The documents listed below are cited as reference information only. They provide examples of equipment that has been found to be acceptable in the past. There is, however, no guarantee that this design will meet all the requirements. Contractors may provide a design of their own that must meet the performance requirements specified herein. The following drawings may be obtained by contacting AMSTA-TR-D/210 (Chitta Das), Warren, MI 48397-5000.

TA13226E1581 - Repair Kit, 6 Inch Hose Line

6.6 Definitions.

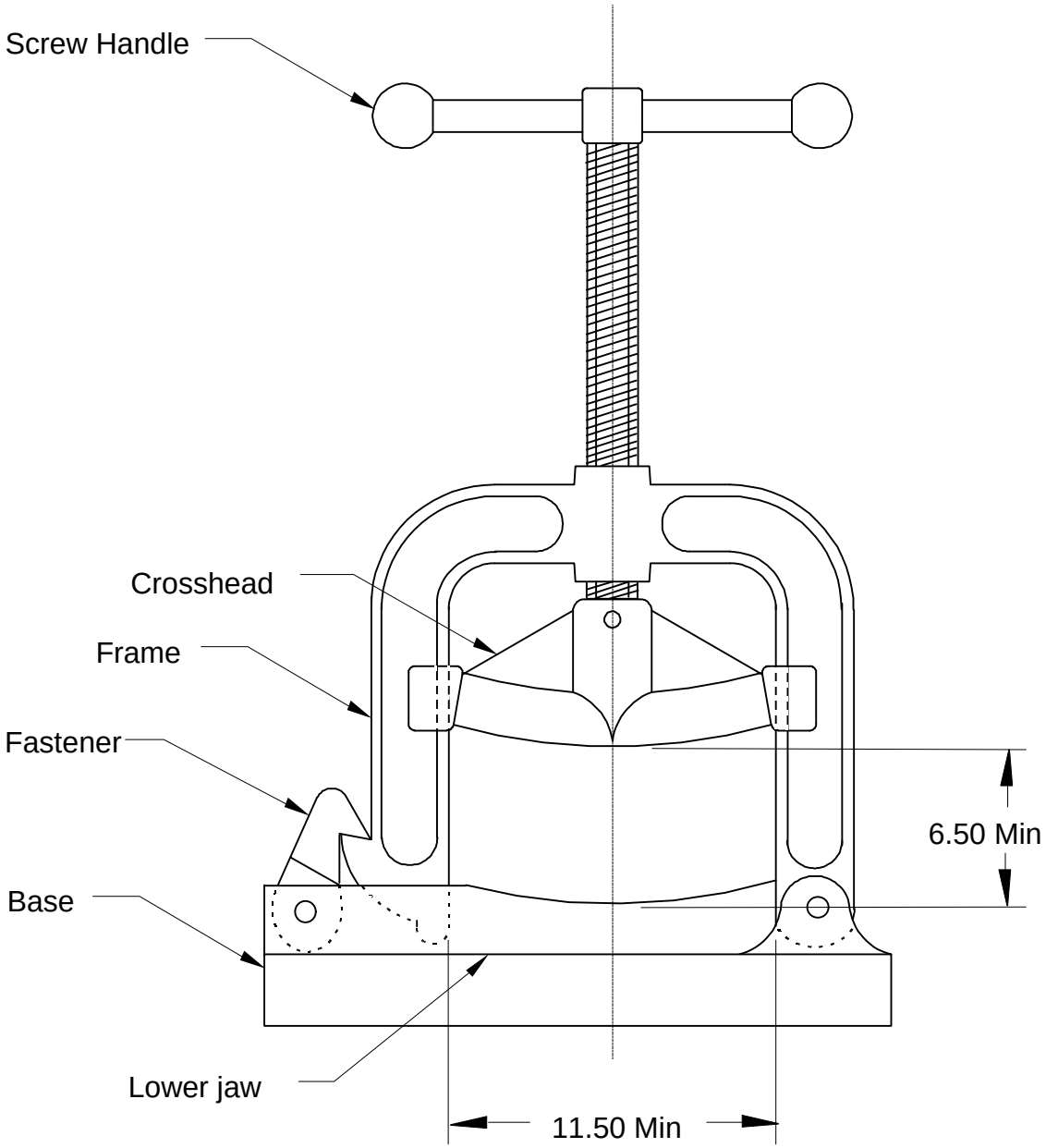
6.6.1 Dissimilar metals. For its purpose, this document considers metals dissimilar when two specimens in contact with each other promote accelerated galvanic corrosion (see 3.2.3.1).

6.6.2 Recovered materials. For the purpose of this document, recovered materials are those materials that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials (see 3.2.2).

6.7 Subject term (key word) listing.

Fuel, diesel
Fuel, aviation
Clamps, hose

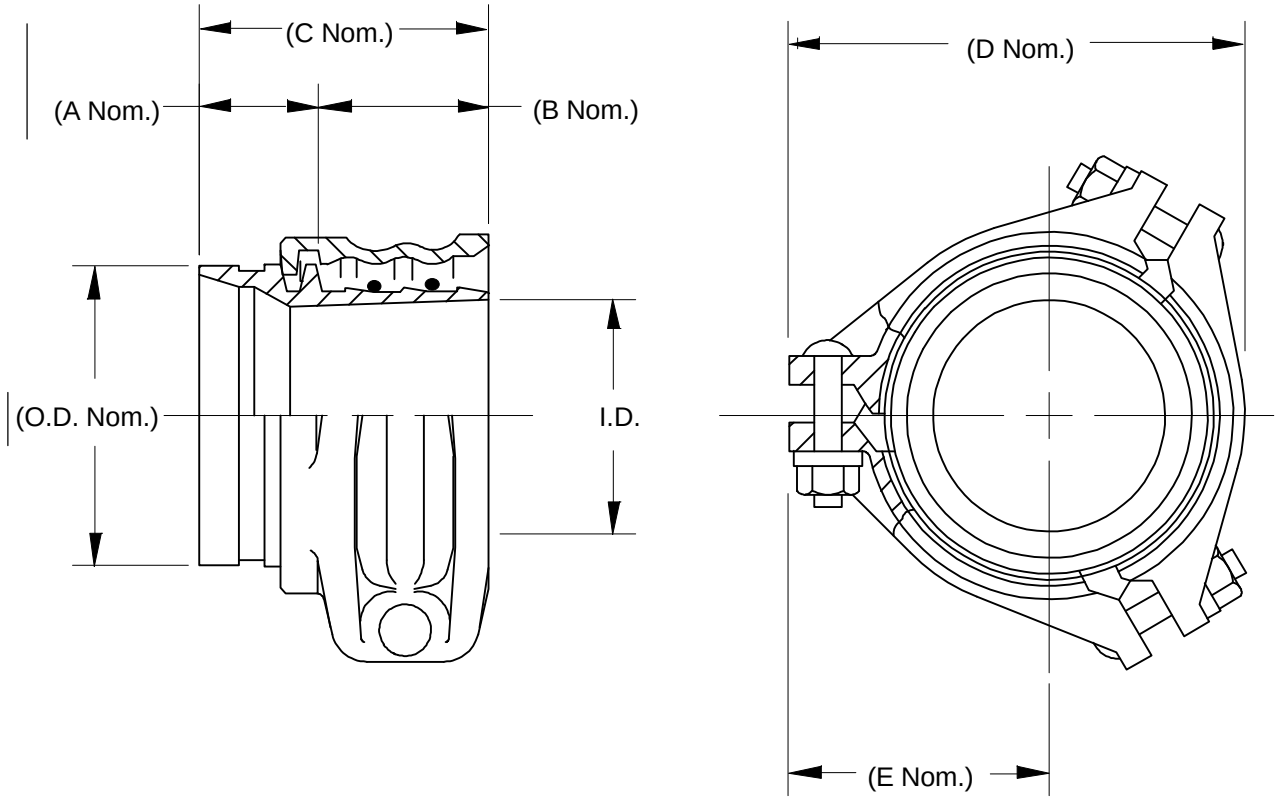
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NOTE: Dimensions are in inches.

FIGURE 1. Hose Clamp - Example.

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Nom. Size	Dimensions							Approx Wgt	Hose Wall Thickness		Hose ID
	A	B	I.D.	O.D.	E	D	C	Lb*	Min	Max	
6.00	1.80	2.70	5.40	6.64	4.90	9.80	4.50	4.50	0.09	0.21	6.00

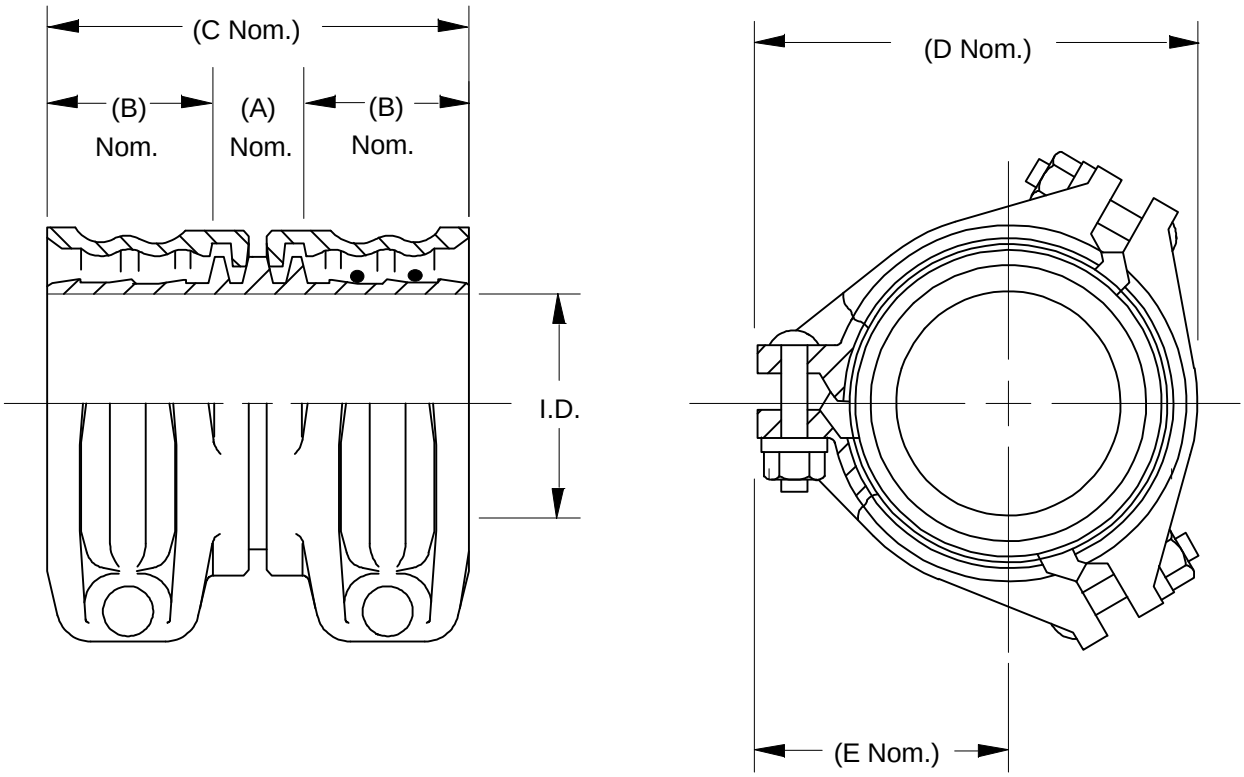
* Nipple and clamp assembly complete.

NOTES:

1. Dimensions are in inches.
2. Tolerances .XX ± .06.

FIGURE 2. Hose Nipple Assembly, Grooved End - Example.

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Nom. Size	Dimensions							Approx Wgt	Hose Wall Thickness		Hose I.D.
	A	B	I.D.	O.D.	E	D	C	Lb*	Min	Max	
6.00	1.40	2.70	5.40	N/A	4.90	9.80	6.80	7.50	0.09	0.21	6.00

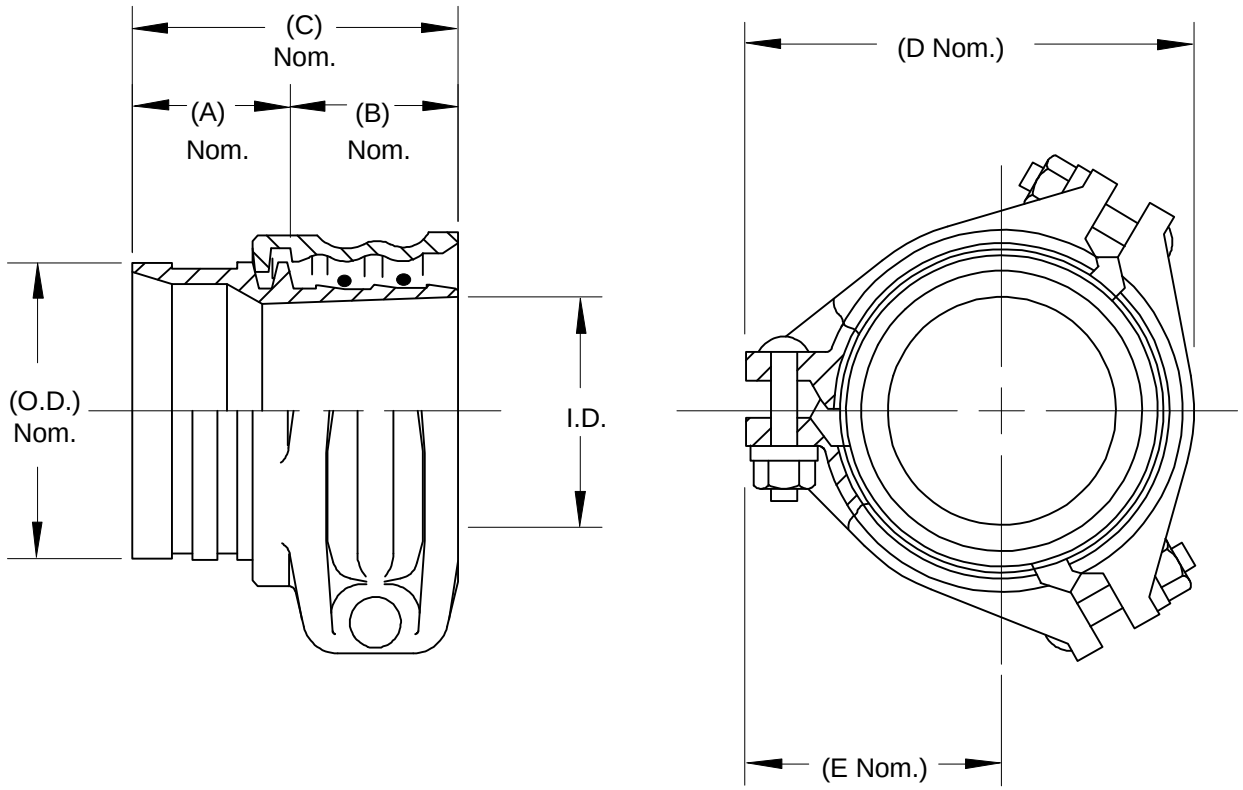
* Nipple and clamp assembly complete.

NOTES:

1. Dimensions are in inches.
2. Tolerances .XX ± .06.

FIGURE 3. Hose Repair Assembly, Grooved End - Example.

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Nom. Size	Dimensions							Approx Wgt	Hose Wall Thickness		Hose I.D.
Inches	A	B	I.D.	O.D.	E	D	C	Lb*	Min	Max	
6.00	2.37	4.19	5.40	6.60	4.90	9.80	6.56	5.00	0.09	0.21	6.00

* Nipple and clamp assembly complete.

NOTE:

1. Dimensions are in inches.
2. Tolerances .XX ± .06.

FIGURE 4. Adapter Assembly, Hose, Double Grooved End - Example.

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INSTRUCTIONS

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-53013B	2. DOCUMENT DATE (YYMMDD) 990201

3. DOCUMENT TITLE

REPAIR KIT, 6-INCH HOSELINE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME USA TACOM, AMSTA-TR-D/210 (FWH)	b. TELEPHONE (Including Area Code) (1) Commercial (810)574-7744 (2) DSN 786-7744
c. ADDRESS (Include Zip Code) US Army Tank-automotive and Armaments Command Attn: AMSTA-TR-D/210 Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 681-9340 DSN 761-9340

DD Form 1426, Oct 89 *Previous editions obsolete*

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